# 1. Clinical Question & Core Hypothesis

## Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

## Working Hypothesis

Primary hypothesis: In tumors with high TIL density and MSH6 deficiency (which yields elevated neoantigen load), systemic HDL-C levels determine the realized efficacy of immune checkpoint blockade by modulating TIL effector function; low HDL-C dampens TIL cytotoxic activity and cytokine production, reducing therapeutic response, whereas normal/high HDL-C preserves TIL function and enhances response. Secondary hypothesis: Three-way stratification by TIL density, MSH6 status, and HDL-C will identify a highly responsive subgroup (TIL-high, MSH6-deficient, HDL-C normal/high) and a discordant subgroup (TIL-high, MSH6-deficient, HDL-C low) despite high neoantigen burden; HDL-C mediates these effects in part through shaping the tumor immune microenvironment via myeloid polarization and T cell metabolism.

| Element | Value |
|---|---|
| Imaging feature | Tumor-Infiltrating Lymphocyte (TIL) density |
| Anchor gene(s) | MSH6, HDL-C |
| Primary endpoint | therapy_response |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**MSH6:** MSH6-associated high neoantigen potential) do not uniformly benefit from immunotherapy—what host- or tumor-extrinsic factors convert immune infiltration into effective anti-tumor responses? 2) Tier 1 paths used: I built on MSH6 (tumor-intrinsic mismatch-repair/neoantigen axis) and HDL-C (host systemic lipid/metabolic state with context-dependent prognostic signals) because these two axes give a clear mechanistic contrast—tumor antigenicity vs host immunometabolism—that can explain variable immunotherapy benefit.

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## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|
| Secondary | MSI | Supporting modifier from knowledge graph |

| | | |
|---|---|---|
| Secondary | HRR | Supporting modifier from knowledge graph |
| Secondary | pro-PSA | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** The interaction between the imaging feature and anchor gene predicts clinical outcome.

**Prediction 2:** Spatial analysis will confirm the mechanistic link at the tissue level.

**Prediction 3:** Functional perturbation will establish causality.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

## 5. Strengths, Caveats & Alternative Directions

### Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

## Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

## Alternative Directions

- MSI: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- HRR: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.